

Model Evaluation

Offline

Assuming we're evaluating the full retrieval redesign including the reranker, the metrics to be used are: **Mean Reciprocal Rank**, **success@k** (fraction of sessions where at least one clicked product appears in the top k positions) , and **recall@k** (for retrieval). The primary assumption here is that success is derived from historic user interaction with the items in the catalog (everything the user clicked on or purchased in a search session).

MRR helps us understand how well positioned relevant items are, **success@k** tells us how if we had a success (relevant item) in our retrieved set, and **recall@k** is a generalization of **success@k** that tells us how many relevant items showed up in our retrieved set out of all items.

Note that offline retrieval evaluation is inherently limited here. We can only measure how well the new retrieval system recovers products that the old system surfaced and users clicked on. Meaning, we should design our online evaluation pipeline to give us a stronger signal that we can rely on as a proxy for performance. Both online and offline evaluation are important here.

While I did not include **nDCG** as a metric in parts 1/2 (because 91% of the session had 1 click only), it becomes relevant when richer engagement signals are available.

Online

There are two key points to cover here: the metrics to optimize for and A/B testing.

Online metric selection should be business-context driven and not easily gamified. A prime example is **revenue per search session** (which requires clearly defining what a session is and recording/logging necessary data) and **conversion rate per session**, **CTR** can be used as a secondary metric, and **zero-click rate** for retrieval.

When it comes to A/B testing, stratified randomization at the user level would be the primary strategy (example dimensions: spending tier and language). This is a hedge against potential power-users that skew behaviour distribution on the Ounass. The key idea is to break down the population and randomly select accordingly as the solution gets rolled out to the A/B groups.

Lastly, I'm assuming luxury e-commerce has lower traffic than other channels so conversion early on is likely to be sparse and it might take some time for statistical significance.